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1 GOAL

This document presents an overview of Link2Valves' public API, its requirements, and its format specification.

2 OVERVIEW

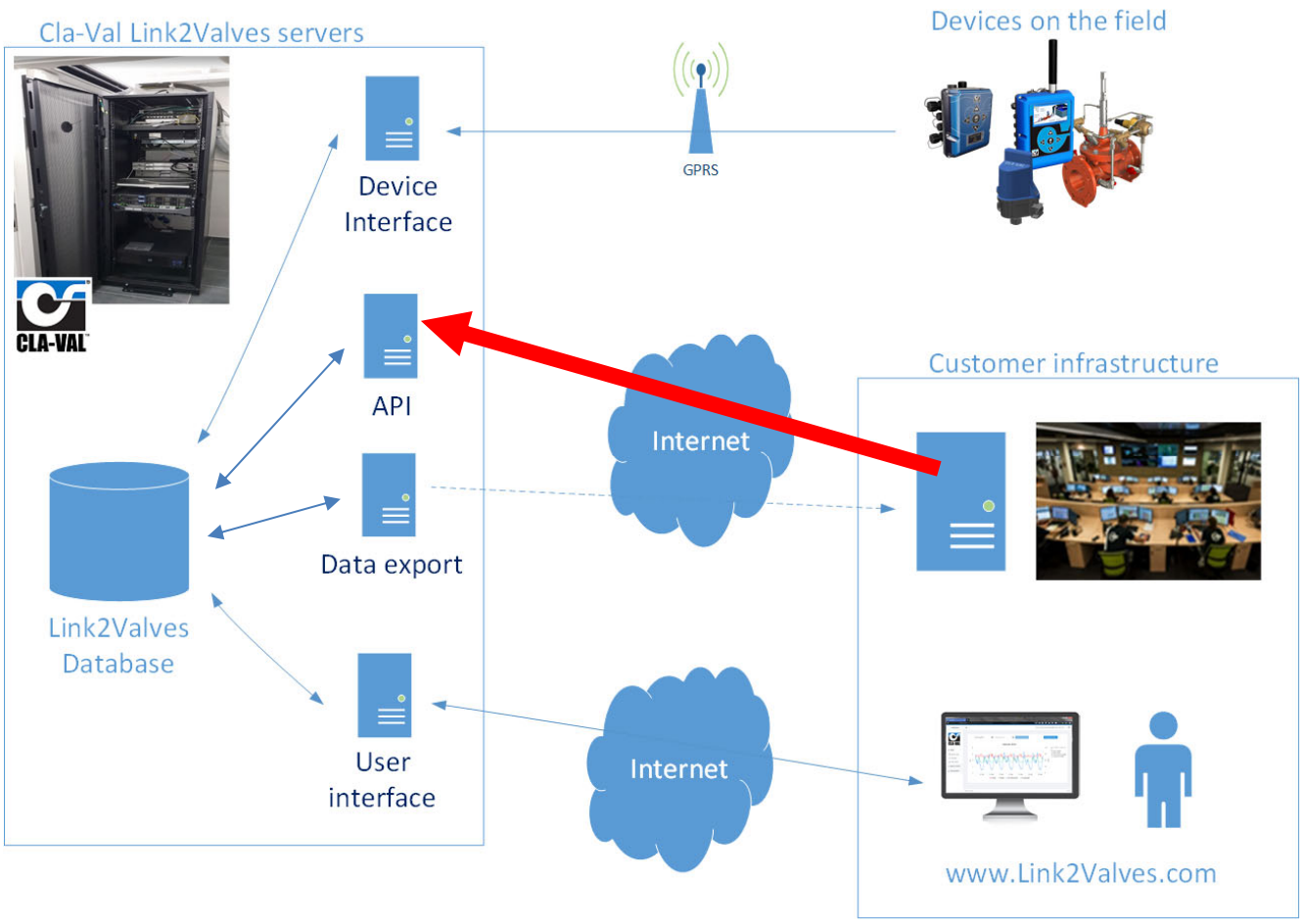
Link2Valves' JSON API allows customer IT infrastructure to seamlessly integrate the functionality provided by the Link2Valves web platform, including:

- Retrieving a Device's data history
- Remotely configuring a Device
- Retrieving Asset Management data such as Valve information

This way, Link2Valves data can be exploited by the customer infrastructure without the need to manually visit the Link2Valves website.

The approach involves making requests using HTTPS and authorized through a client certificate, which enforces strong, state-of-the-art Internet security standards.

An indication of the role of the public API inside the Link2Valves ecosystem is shown by the red arrow in the figure below:



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3 ADVANTAGES

- Data retrieval at any moment, without relying on fixed intervals
- Remote configuration managed within the customer's existing IT interfaces
- Customization of the request parameters at the customer convenience, without need for intervention by CLA-VAL
- Easy parsing of the responses by leveraging the standard JSON format
- Secure, seamless integration into existing IT infrastructure by leveraging the ubiquitous HTTPS protocol
- Requests being outgoing only; no need for the customer to open their IT environment to receive any kind of connection by CLA-VAL

4 REQUIREMENTS

4.1 INFRASTRUCTURE REQUIREMENTS

The customer is responsible of developing the business logic that:

- Will regularly launch the HTTP requests, integrating the client certificate generated by CLA-VAL.
- Will interpret the JSON result.
- Will save the result, back it up, process or otherwise consume the data inside the customer's infrastructure.

The customer will also need to follow a simple procedure to generate its own private client certificate. CLA-VAL will provide assistance during the process.

4.2 SECURITY REQUIREMENTS

Requests must use HTTPS.

Authorization is achieved through the combination of the following:

- The client certificate provided in the request.
- A Link2valves user, created by CLA-VAL on request, associated with the provided client certificate.
- The assignment to the Link2valves user of permissions allowing access to the requested devices.

CLA-VAL can, on request, provide assistance to put in place any of these authorization requirements.

5 REQUESTS

5.1 GET LIST OF DEVICES

https://www.link2valves.com/public/api/devices

Purpose: Retrieving the list of the Devices in Link2Valves that are accessible by the client certificate used in the request. Useful metadata is contained in each Device object, including the unique IMEI number that identifies it in other requests.

Method: GET

Response: An array of **Device** objects. The Channel objects in this response will **not** contain any TimeValues.

5.2 GET DEVICE

https://www.link2valves.com/public/api/devices/<device_imei>

Purpose: Retrieve metadata for a single Device, such as its name and its configuration.

Method: GET

Response: A single **Device** object. The Channel objects in this response will **not** contain any TimeValues.

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5.3 GET DEVICE DATA

https://www.link2valves.com/public/api/devices/<device_imei>/data

Purpose: Retrieve metadata and a history of measurements (TimeValue data points) from a single device.

Method: GET

Parameters:

Parameter	Type	Required	Description
periodStart	UE timestamp in millisecs	No	Beginning of the period for which data is requested. If not present, the API's current allowed minimum is assumed.
periodEnd	UE timestamp in millisecs	No	End of the period for which data is requested. If not present, the current time is assumed.
dataTypes	String or Array of String	No	Types of channels (data lines) to be included in the response. If not present, all available types are assumed. Available types are: • pressureIn • pressureOut • flow • Set Point • Valve Position • e-Drive CMD • e-Drive FB • Vbatt • pressureTarget • flowTarget • volume
chanNames	String or Array of String	No	Names of the channels (data lines) to be included in the response. Typically, dataTypes will be left null when using chanNames, but both properties work in addition if both are specified.

Response: A single **Device** object, complete with TimeValues for the requested channels/period in the Device's *channels* property.

5.4 POST CONFIG

https://www.link2valves.com/public/api/devices/<device_imei>/config

Purpose: Sending configuration changes to a target Device. The changes will either be fully accepted, resulting in HTTP code 200, or fully rejected with code 422. Configuration will never be applied partially.

Method: POST

Parameters (Body):

Parameter	Type	Required	Description
deviceConfig	DeviceConfig object	Yes	Configuration changes to be applied to a target Device at the earliest opportunity.

Response: A single **ConfigResult** object.

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6 OBJECT DESCRIPTION

DEVICE			
Information related to a single device			
Property	**Type**	**Required**	**Description**
id	Integer	Yes	Per-device unique identifier. Device ids are assumed to refer to CLA-VAL-generated ids unless the *idOrigin* field is explicitly specified
imei	Number	No	Unique IMEI number associated to the Device's hardware
idOrigin	String	No	Name of the entity that generated the id given in the *id* field **Note:** If not present, CLA-VAL is assumed
modulationDeviceId	Integer	No	Id of the CLA-VAL device attached to the modulating valve of the hydraulic network Intended for Critical Points to indicate their dependency on a particular modulating valve
name	String	No	Human-readable name given to the device
settings	Array of Setting objects	Yes	List of configuration options, their details, and their current values
channels	Array of Channel objects	Yes	Relevant inputs and outputs of the device, their units and range
curves	Array of Curve objects	No	The Control Curves currently applied in the device
timeZone	String	No	IANA timezone id string, indicating local time in the device. Useful to contextualize time-based Control Curves **Note:** If not present, "Etc/UTC" is assumed
valve	Valve object	No	The valve associated with this device
site	Site object	No	The site in which this device is located

VALVE			
A physical hydraulic valve			
Property	**Type**	**Required**	**Description**
id	Integer	Yes	Unique identifier of the valve, generated by CLA-VAL
name	String	Yes	Name of the valve
brand	String	No	Brand of the valve
partNumber	String	No	Part number (also called Serial number in some languages)
intModelNumber	String	No	Model number of the valve
size	String	No	Size of the valve. Usually given as PN (40, 16, 10…) or making its units explicit
installed	Seconds from Unix Epoch	No	Date of installation (commissioning) of the valve
tags	Array of strings	No	User-defined custom tags assigned by the valve. For example, "outdoors", "maintenance contract "2719638" or "reservoir"
customFields	Array of Custom Field objects		Custom, User-defined data fields assigned to the valve
notes	String	No	Text note assigned to the valve

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CUSTOM FIELD			
A custom data field, consisting of a name and its associated value			
Property	**Type**	**Required**	**Description**
name	String	Yes	The name of the field. For example: "asset number"
value	String	No	The value of the field. For example: "14627B"

SITE			
A physical location where a valve or a device is installed			
Property	**Type**	**Required**	**Description**
id	Integer	Yes	Unique identifier of the Site, generated by CLA-VAL
name	String	Yes	User-defined name given to the Site
addressField	String	No	Text address of the field. For example, "1701 Placentia Ave, Costa Mesa"
latitude	Number	No	Latitude coordinate of the Site's location
longitude	Number	No	Longitude coordinate of the Site's location
customFields	Array of Custom Field objects	No	Custom, User-defined data fields assigned to the Site in Link2Valves

SETTING			
Configuration variable of a device			
Property	**Type**	**Required**	**Description**
name	String	Yes	Unique name for this configuration option
value	Number	Yes	Configured value for this setting
units	String	No	Units in which the *value* is expressed
min	Number	No	Minimum acceptable value
max	Number	No	Maximum acceptable value

CHANNEL			
Input, output or other variable of a device. When of type input, it may contain a series of measurements.			
Property	**Type**	**Required**	**Description**
port	String	Yes	Short identifier for this channel. Corresponds to the ports shown in the physical device's menus. Examples: *AI3*, *AO1*, *DI2_Flow*, *SO1*, *CPC SP*
name	String	Yes	Human-readable name for this channel
type	String	Yes	The physical role of its values. Possible types are: • pressureIn • pressureOut • flow • Set Point • Valve Position • e-Drive CMD • e-Drive FB • Vbatt • pressureTarget • flowTarget • volume
units	String	Yes	Units in which its values are measured. Unless other units

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			are configured in the controller generating the data, typical units are: • m3/s • m3/h • L/s • L/min • ft3/s • ft3/min • Mgal/day • gal/min • ML/day • UK gal/min • m3 • L • ML • gal • UK gal • ft3 • Mgal • UK Mgal • Pa • bar • kPa • mH2O • lbf/in2 • s • min • h • day • V
min	Number	No	Minimum acceptable value
max	Number	No	Maximum acceptable value
timeValues	Array of TimeValue objects	No	Latest timeseries data generated by this channel. This data will only be included in the response when the requests has been made with *data* as its *purpose* parameter

TIMEVALUE			
A single timeseries data point			
Property	**Type**	**Required**	**Description**
time	Seconds from Unix Epoch	Yes	Time coordinate of the data point
value	Number	Yes	Value of the data point at the given time

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CURVE			
One of the Control Curves (profiles) applied in a device			
Property	**Type**	**Required**	**Description**
name	String	No	Human-readable name for this curve
appliedFrom	Seconds from Unix Epoch	No	Moment in time when this curve was applied by the device
shape	String: “slope” or “step”	Yes	• **Slope:** the curve is formed by joining the given points with straight lines • **Step:** the curve is formed by keeping a point’s value until the next data point, where it immediately jumps to and keeps the next value **Note:** Curves with an "activeCurve" axis (see below) must have "step" shape **Note:** The last "step" is determined by the two last data points, which must have the same value
xAxis	Axis object	Yes	Definition of the input axis of the control curve
yAxis	Axis object	Yes	Definition of the output axis of the control curve
points	Array of Point objects	Yes	A minimum of 2 Points that conform the curve
externalId	Integer	No	Id given to this curve for external identification purposes

AXIS			
Defines one of the axis of a curve. Usually, corresponds to a Channel in the target Device.			
Property	**Type**	**Required**	**Description**
type	String	Yes	The physical role of its values, such as "flow" or "pressureOut" The following special values are also possible: • ***timeOfDay*:** Specifies an amount of time from 00:00 every day. For example, if the units are minutes, 16:15 corresponds to the value 111 • ***dayofWeek*:** Specifies a day of the week, from 0 for Monday to 6 for Sunday • ***activeCurve*:** Specifies the schedule of activation of other control curves. The units of this type of axis must be “*curveIndex*”
units	String	Yes	Units in which its values are measured **Note:** For an "*activeCurve"* axis type, the units must be "*curveIndex"*, corresponding to the position (starting at 0) of the target control curve in the device’s array of curves
channel	String	No	The name of the channel (data line) in the Device that is associated to this axis of the control curve, if available. For example, “Flow”. Note that this channel may not always be determined.

POINT			
A data point that defines a curve			
Property	**Type**	**Required**	**Description**
x	Number	Yes	Coordinate for the x (input) axis
y	Number	Yes	Coordinate for the y (output) axis

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DEVICECONFIG			
Defines configuration changes to be applied on a device			
Property	**Type**	**Required**	**Description**
settings	Array of Setting objects	No	List of one or more settings in the device, containing the new *value* to be applied to each of them. If this property is missing or *null*, no changes will be applied to the target Device's existing settings
curves	Array of Curve objects	No	The control curves to be applied by the target Device, replacing any previously existing ones. If this property is missing or *null*, no changes will be applied to the target Device's existing curves. If an empty array is given, the curves in the target Device will be deleted

CONFIGRESULT			
Response to a configuration change request, containing any generated success or error messages			
Property	**Type**	**Required**	**Description**
accepted	Boolean	Yes	Whether the requested changes were valid and accepted for application to the Device at the earliest opportunity
errors	Array of Strings	No	When *accepted* is *false*, this array may contain one or more strings describing the problems with the requested changes that made the server reject the configuration
eta	Integer	No	When *accepted* is *true*, estimated delay in minutes before a Device will be able to apply the configuration changes requested. This value is based on the target Device's communication period but may be influenced by other factors and thus it is just an estimate for information purposes

7 EXAMPLE CONFIGURATION WORKFLOW

1. Review the list of Devices to view their current configuration

GET https://www.link2valves.com/public/api/devices


```
[
  {
    "id": 120,
    "imei": 999917050003077,
    "name": "D22-BARCELONE-REGULATION",
    //...
    "settings": [
      {
        "name": "Full_Opening",
        "value": 1.0000,
        "units": "",
        "min": 0.0000,
        "max": 1.0000
      },
      {
        "name": "LSP_ON",
        "value": 1.0000,
```


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```
    "units": "",
    "min": 0.0000,
    "max": 1.0000
  },
  {
    "name": "Mode",
    "value": 1.0000,
    "units": "0:Auto,1:Manu",
    "min": 0.0000,
    "max": 1.0000
  },
  {
    "name": "Remote SP",
    "value": 3.2000,
    "units": "bar",
    "min": 0.0000,
    "max": 16.0000
  }
  //...
],
"channels": [
  {
    "port": "AI1",
    "type": "pressureIn",
    "name": "P1",
    "units": bar,
    "min": 0.0000,
    "max": 16.0000,
    "timeValues": []
  },
  {
    "port": "AI2",
    "type": "pressureOut",
    "name": "P2",
    "units": "bar",
    "min": 0.0000,
    "max": 16.0000,
    "timeValues": []
  }
  //...
],
"curves": [
  {
    "name": "Target_P vs Time",
    "externalId": null,
    "appliedFrom": 1622540712404,
```


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```
      "shape": "step",
      "xAxis": {
        "type": "timeOfDay",
        "units": "min"
      },
      "yAxis": {
        "type": "pressureTarget",
        "units": "mH2O"
      },
      "points": [
        {
          "x": 0.00000,
          "y": 5.50000
        },
        {
          "x": 360.00000,
          "y": 6.70000
        },
        {
          "x": 1320.00000,
          "y": 5.50000
        }
      ]
    }
  ],
  "timeZone": "Europe/Berlin",
  "valve": {
        "id": 21,
        "name": null,
        "brand": "",
        "intModelNumber": "D12-99-01",
        "size": "DN 80",
        "installDate": "2017-09-14T22:00:00",
        "tags": [
          “sector 13”,
          “important”,
          “2017”
        ],
        "notes": null,
        "customFields": [
            {
                “name”: “contract type”,
                “value”: “5-year maintenance all inclusive”
            },
            {
                “name”: “asset id”,
```


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```
                "value": "831267"
              }
            ]
          },
          "site": {
            "id": 43,
            "name": "Demo_Device",
            "addressField": "Chemin des Mésanges 1, 1032 Romanel-sur-Lausanne, Switzerland",
            "latitude": 46.5682979000,
            "longitude": 6.6027804999,
            "customFields": [
              {
                "name": "zip code",
                "value": "CH1032"
              }
            ]
          }
        },
        {
          "id": 131,
          "imei": 999917050003098,
          //...
        }
      ]
```


2. Find out the IMEI number of the Device to be configured and the configuration changes to be applied. For example:
   Device to configure: D22-BARCELONE-REGULATION
   IMEI: 999917050003077
   Changes to settings:
   - Activate Auto Mode
   - Deactivate Local Set Point
   - Add a point to the Control Curve so that the pressure at 8:30 am is set to 7.2 bar
3. Send the configuration changes to the API
   POST https://www.link2valves.com/public/api/devices/999917050003077/config


```
{
  "settings": [
    {
    "name": "LSP_ON",
    "value": 0
    },
    {
    "name": "Mode",
    "value": 0
    }
  ],
  "curves": [
    {
```


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```
    "name": "Target_P vs Time",
    "shape": "step",
    "xAxis": {
      "type": "timeOfDay",
      "units": "min"
    },
    "yAxis": {
      "type": "pressureTarget",
      "units": "mH2O"
    },
    "points": [
      {
      "x": 0,
      "y": 5.5
      },
      {
      "x": 360,
      "y": 6.7
      },
      {
      "x": 510,
      "y": 7.2
      },
      {
      "x": 1320,
      "y": 5.5
      }
    ]
    }
  ]
}
```


4. The response to the configuration request confirms that the changes are accepted and that they will be applied by the Device in around 43 minutes.


```
{
  "accepted": true,
  "errors": [],
  "eta": 43
}
```


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